Name: ______ Cohort/Batch: ______ Date: ______ Score: ______

OPC-UA PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A IoT & Edge

TOTAL: 10 QUESTIONS

Q1 What is OPC-UA and what problem in IoT & Edge does it solve? *Threat Model*

Q2 What are the core components or concepts in OPC-UA? *Architecture*

Q3 How does OPC-UA compare to its main alternatives? *Configuration*

Q4 How do you get started with OPC-UA for a new project? *Hardening*

Q5 What configuration options most affect OPC-UA's behavior in production? *SSO / IdP*

Q6 How does OPC-UA handle reliability and high availability? *Observability*

Q7 What observability does OPC-UA provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in OPC-UA? *Compliance*

Q9 What security best practices apply when running OPC-UA? *Testing*

Q10 What mistakes do teams commonly make when adopting OPC-UA? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 OPC-UA is a tool or platform that

Mitigation

OPC-UA is a tool or platform that automates or simplifies a specific concern in the IoT & Edge domain, reducing manual effort and operational complexity for engineering teams.

A2 OPC-UA has key building blocks that work

Primitives

OPC-UA has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 OPC-UA trades off simplicity against feature richness

Hardening

OPC-UA trades off simplicity against feature richness differently than its alternatives. Choose OPC-UA when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install OPC-UA via its package manager or

Gotchas

Install OPC-UA via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include concurrency

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 OPC-UA provides HA through leader election, replication,

Observability

OPC-UA provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 OPC-UA exposes metrics (Prometheus endpoint or built-in

Automation

OPC-UA exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from misconfigured

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run OPC-UA with the principle of least

Assessment

Run OPC-UA with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the documentation and

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.